Evidential Fusion for Calibrated Edge Traversability on a Wheeled Delivery Robot

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Abstract

A sidewalk delivery robot follows a route from a map, but maps go stale: a path marked as a walkway may contain stairs or a curb that a wheeled robot cannot cross. Camera and LiDAR can catch the error, but each fails in its own way. This project combines camera, LiDAR, and map into one *confidence* that a path segment is safe, which also reports its uncertainty and when the sources disagree. The robot crosses only on trustworthy confidence, looks closer when unsure, and reroutes around hazards. Evaluation replays real campus robot logs with injected sensor faults and stale maps.

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1 Problem and Decision Boundary

A wheeled delivery robot follows a campus sidewalk graph whose edge labels come from a map prior (OpenStreetMap OpenStreetMap contributors [2026]). Labels can be stale: an edge tagged `footway` may contain stairs or a curb.

Decision: for each edge e on the route, may the robot commit to traversing it?

- **Act:** traverse e.
- **Defer:** an obligation is unresolved and closer observations remain; re-observe.
- **Repair:** an obligation fails; remove e, replan, record a map patch.
- **Handoff:** no admissible route remains; escalate to an operator.

Admitting a curb or stairs is catastrophic, while a false rejection costs a detour, so the gate is fail-closed.

Research question: does extending evidential multi-view fusion (TMC Han et al. [2023]) to discount the map's reliability by label age and accumulate evidence over the approach give better-calibrated edge confidence than softmax fusion and standard TMC, and can a gate on it match a "reject if any source objects" veto on false admission with fewer unnecessary rejections? Perception is out of scope: views are frozen, cached features.

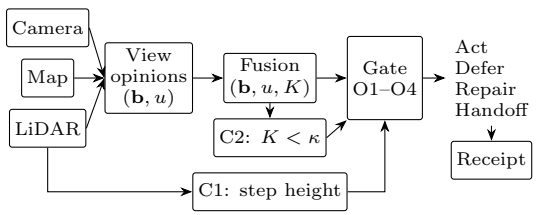


Figure 1: Per-view opinions are fused, checked by C1 and C2, and gated; every decision is logged to the receipt.

2 Modalities, Data, Typed Claims, Risks, and Obligations

Data: *GND* Liang et al. [2025] (CC0), which includes a labeled traversability map. Training uses UMD map 3 (McKeldin Mall), split by location; Georgetown is held out. Bags hold LiDAR, RGB, odometry, and SLAM poses, and are cached once, so no network is needed afterward. Edges are labeled *traversable* or *hazard* (GND's stairs-or-curb class); a curb and stairs both mean a reroute.

Typed claims: each view turns its feature into an opinion $(\mathbf{b}, u)$ over the two classes, and the edge is classified from the *fused* opinion (Table 1). The map is never certain: its opinion is discounted by label age, $\mathbf{b}' = r\mathbf{b}$, $u' = 1 - r(1-u)$, with $r = r_{\max} e^{-\mathrm{age}/\tau_a}$ and $r_{\max} < 1$, so an old label adds uncertainty, not hazard belief. Every claim records edge, value, uncertainty, source, tier, and provenance (sensor, timestamp, range, artifact hash, version).

Risk tiers set how much evidence Act needs, from OSM context known before arrival: **T0** interior walkway; **T1** meets a roadway or crossing (likely curb); **T2** within d m of mapped steps or kerbs. An edge escalates to T2 if its fused hazard belief ever reaches β.

Obligations: any FAIL ⇒ Repair; any UNRESOLVED ⇒ Defer, or Repair if no closer evidence remains.

- **O1** (all): view conflict $K < \kappa$ (C2).
- **O2** (T1+): max step < 0.05 m with enough points (C1); for T2, on ≥ 2 frames at different ranges.
- **O3** (all): fused $b_{\mathrm{trav}} \geq \tau_{\mathrm{tier}}$ and $u \leq \tau_u$.
- **O4** (all): every claim resolves to a hashed artifact.

Risks: few hazard sites (counted first); OSM edit time is not survey time; Dempster's rule hides stale-map conflict; C1 and the LiDAR view share a sensor.

3 CPU Baseline: Estimated Runtime and Memory

Estimates on 4 cores and 16 GB for ∼3.7k frames at 2 Hz; measured values go to `compute.json`.

Source (frozen feature)	Example claim	Role
Camera (DINOv2 Oquab et al. [2024] embedding)	`terrain(e): b=(.15,.70), u=.15`	Semantic evidence (A1)
LiDAR (step-height histogram)	`terrain(e): b=(.05,.81), u=.14`	Geometric evidence
Map prior (OSM tag, 5-year-old label)	`terrain(e): b=(.58,0), u=.42`	Age-discounted prior
Fusion (TMC / proposed)	`b=(.12,.85), u=.02, K=.47`	Fused opinion (A2)
C1 step height (raw points)	`max_step(e)=0.14 m, n=812`	Executable checker

Table 1: Example typed claims on a stale edge mapped as `footway` that has stairs; $\mathbf{b}$ = (traversable, hazard). K is the largest conflict between view opinions, used by checker C2.

Stage	Time	RAM
Bags (72 GB) → 2 Hz cache, once	65 min	<2 GB
Graph, association, labels	<5 min	<2 GB
Frozen features, C1	10 min	3 GB
Fusion ladder (9 methods × 5 seeds)	<10 min	<2 GB
Gate and receipts	20 min	<3 GB
Baseline / evaluator	1.8 h / 30 min	<6 GB

Both fit the 4 h / 2 h budgets. **Minimum fallback:** LiDAR and map views with fixed Dempster–Shafer masses Valente et al. [2018], plus C1 and C2.

4 Checkers, Action Gate, Invariant, and Failure Injection

C1, geometric step height: fits a ground plane to raw LiDAR points on the edge (RANSAC Fischler and Bolles [1981]) and reports the largest step. No learning.

C2, fusion conflict: reports the largest Dempster conflict between pairs of view opinions. It reads only opinions, never geometry, and flags the disagreement a stale map creates.

Gate: a deterministic function of claims and tier. Missing evidence is UNRESOLVED, never PASS. Deferral consumes the next, closer cached frame.

Invariant I1: no edge is admitted unless every obligation for its tier passes. It is checked on every receipt and by property-based tests.

Receipt: config and data hashes, seed, faults, and per-edge claims, obligations, and decision.

Failure injection (expected detection in parentheses):

- *Stale map:* stair and curb edges flipped to `footway` with assigned label ages (C2 conflict).
- *Adversarial:* stale map plus flat-sidewalk crops swapped in for stairs (only the LiDAR view and C1 remain).

Unwarranted claims: none about real-world safety, other campuses, hazards beyond steps and stairs, or physical causation.

5 Metrics, Baselines, Ownership, Timeline, and Fallback

Task performance: ECE, NLL, hazard F1, AUROC for stale-edge detection, and risk–coverage.

Fusion baselines: map only; single view; early, late, and attention fusion (A2); fixed Dempster–Shafer masses Valente et al. [2018]; TMC Han et al. [2023] and QMF Zhang et al. [2023]; proposed.

Policy behavior: false admission (Wilson CI), unnecessary rejection and deferral, checker coverage, failure detection, repair success (goal reached using only truly traversable edges; detour overhead vs. an oracle map), runtime, and peak memory. Policies: map-only route, conservative veto, gate on TMC, gate on proposed fusion.

A1/A2/A3: cached embeddings (A1); evidential opinions with conformal thresholds Angelopoulos and Bates [2021] (A2); per-decision attribution by ablating each view (A3). **Ownership:** solo.

Timeline: *Nov 7–20:* cache, labels, features, C1, baselines. *Nov 21–30:* C2, gate, receipts, one injection. **Dec 1 progress gate.** *Dec 2–10:* proposed fusion, all injections, A3, poster. *Dec 12–15:* report.

Fallbacks: proposed loses to TMC ⇒ gate on TMC; UMD alignment or association fails ⇒ Georgetown becomes primary.

Repository: `https://github.com/nalddox/ml4ss-terrain-gate`. **Data access:** GND UMD map 3 north (chunks 01–15), Georgetown, and both traversability maps are downloaded locally. **Optional scale-up:** none requested; everything runs on CPU.

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